

Milindu Weerawarna

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Profile

Computer Science and Engineering undergraduate at the University of Moratuwa with a strong interest in Cybersecurity and Software Engineering. Experienced in designing and developing secure, scalable software systems across web, cloud, and AI-driven applications. Skilled in secure application development, vulnerability assessment, system design, and modern software engineering practices. Proven problem-solving, analytical, and teamwork abilities demonstrated through cybersecurity competitions, CTF challenges, and multidisciplinary engineering projects.

Education

B.Sc. (Hons) in Computer Science and Engineering Mar 2024 – Present

University of Moratuwa, Sri Lanka | *Cybersecurity Specialization*

- GPA: 3.69 / 4.00 | Dean's List – Semester 1 | Expected graduation: 2028

G.C.E. Advanced Level – Physical Science Stream 2022

Maliyadeva College, Kurunegala

- 3 A passes | Z-Score: 2.7093 | District Rank: 5 | Island Rank: 78

G.C.E. Ordinary Level 2018

Maliyadeva Adarsha College, Kurunegala

- 9 A passes

Projects

ExfilTrack – USB Data Exfiltration Detection Platform ongoing | [GitHub](#)

- Developing a digital forensics platform to detect and investigate potential USB-based data exfiltration incidents on Windows systems
- Correlating Windows Registry, Event Log (EVTX), and LNK/Jump List artifacts to reconstruct device activity timelines and identify suspicious file access patterns
- Implementing automated risk scoring, SHA-256 evidence verification, and reproducible forensic report generation for incident investigations

Technologies: Python, Digital Forensics, Cybersecurity, Windows Forensics, SQLite, SHA-256

Benthic Guardian – AI-Enabled Coral Reef Monitoring 2026 | [Frontend](#) [Backend](#) [LLM Model](#)

- Developed an affordable real-time coral bleaching monitoring and early-warning system; selected as a finalist at the **SLIoT Challenge 2026**
- Integrated underwater sensor nodes, real-time data ingestion, interactive dashboards, and a Physics-Informed Neural Network for bleaching-risk estimation
- Designed a sustainable architecture using microbial fuel cells, solar power, and an underwater optical communication link

Technologies: ESP32, IoT, Python, TensorFlow, PINN, FastAPI, Next.js, PostgreSQL

Disaster Management System 2026 | [GitHub](#)

- Built a real-time platform for monitoring flood and landslide risks through IoT sensor data and live event streaming
- Implemented secure authentication and role-based access for administrators, emergency responders, and public users
- Delivered instant alerts and incident updates using Apache Kafka and Socket.IO

Technologies: React, Node.js, Apache Kafka, Socket.IO, Supabase, IoT

BrightBuy – Full-Stack E-Commerce Platform 2025 | [GitHub](#)

- Developed a full-stack e-commerce platform with secure REST APIs, dynamic product variants, and a persistent shopping cart
- Implemented JWT authentication, role-based access control, stock validation, and bcrypt password hashing

Technologies: React, Node.js, Express.js, MySQL, JWT

- Rebuilt and extended an open-source MCP server for authorized infrastructure reconnaissance and security posture analysis
- Integrated DNS, WHOIS, SSL/TLS, subdomain, ASN, and cloud-exposure analysis with structured reporting and automated tests

Technologies: Python, FastMCP, Pytest, AsyncIO, REST APIs

Skills

Languages:	Python, TypeScript, JavaScript, Java, C, C++, SQL, Bash, VHDL
Frontend:	React, Next.js, Flutter, Tailwind CSS
Backend:	Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets
Cloud/DevOps:	AWS, Google Cloud, Docker, Git, GitHub, Linux, Kubernetes, Jenkins, Ansible, Terraform
Databases:	PostgreSQL, MySQL, MongoDB, Firebase, DynamoDB, SQLite, Redis
AI/ML:	TensorFlow, Scikit-Learn, XGBoost, Pandas, Jupyter,
Cybersecurity:	Burp Suite, Wireshark, Nmap, Nessus, OWASP ZAP, Metasploit, Hashcat

Certifications

• Ethical Hacker – Cisco	Certificate
• Introduction to Cybersecurity – Cisco	Certificate
• API Security Fundamentals – APIsec University	Certificate
• Certified LLM Security Expert (CLLMSE) – Red Team Leaders	Certificate
• KodeKloud Engineer - 100 Days of DevOps KodeKloud – KodeKloud	Certificate
• KodeKloud Engineer - 100 Days of Cloud (AWS) KodeKloud – KodeKloud	Certificate
• Fundamentals of Accelerated Data Science – NVIDIA	Certificate
• Machine Learning Specialization – Stanford University & DeepLearning.AI	Certificate

Competitions & Achievements

• Finalist – SLIoT Challenge (Open Category) Organized by the University of Moratuwa	2026
• 1st runner up – Vectra 8 hour king of the hill Competition Organized by IEEE WIE AG of IIT	2026
• 4th Place – Enigma Mathematical Hackathon Organized by the Mathematical Society, University of Moratuwa	2025
• Finalist – ANIMUS 1.0 Capture the Flag Competition Organized by the Computer Science Students' Community, University of Ruhuna	2025
• Finalist – SLIIT Xtreme Competitive Programming Competition Organized by SLIIT	2025
• Participant – IEEEExtreme Programming Competition Organized by the Institute of Electrical and Electronics Engineers (IEEE)	2025
• 4th Place – Capture the Flag Challenge Organized by the Cyber Security Community of SLIIT	2026

Extracurricular & Volunteering

• Company Coordinator , CSE Career Fair	2026
• Organizing Committee Member , ADScAI Conference	2026
• Teaching assistant and examination supervisor at a higher education institute	
• Chess player and actively participating in international tournaments	

References

Dr. Sunimal Rathnayake
Senior Lecturer
Department of Computer Science and Engineering
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